

Apple Cleaning Systems Master System

Phase 0 Progress Report

Prepared for Alfonso Cazorla, Owner, and Luis Cazorla, Operations Manager

Prepared by ACS Master System Project Team

Date: May 21, 2026

Project: ACS Workflow Master System

Purpose of this report

This document explains the work completed so far in plain business language. It is intended for company leadership and operations management, not software engineers.

Phase 0 is foundation work. It prepares the system for safe future operations before real dispatch actions, vendor integration, production deployment, and employee permissions are added.

Executive Summary

The ACS Master System is being built as a long-term Field Service Management platform for Apple Cleaning Systems. The goal is to replace fragile manual and disconnected workflows with one reliable system that can support dispatch, office operations, Water Emergency work, technician coordination, audit history, future integrations, and future reporting.

So far, the project has completed the foundation stage called Phase 0. This is not the finished operating system yet. Phase 0 creates the safe technical structure needed before real office actions, dispatch actions, vendor integrations, authentication, and production deployment are added.

The project documentation records 34 Phase 0 modules completed or prepared through review. Modules 1 through 33 are committed in the repository. Module 34 is currently review-ready but not committed yet.

The most important achievement is that the system now has a clear foundation for:

- receiving and checking job information
- stopping uncertain work for Manual Review
- keeping Water Emergency work separate from standard jobs
- preparing dispatch and route information safely
- preserving operational history and audit evidence
- showing read-only office dashboard information
- verifying the dashboard against a local PostgreSQL database
- presenting Water Emergency details, visits, equipment context, reviews, blockers, and alerts without giving the screen authority to take action

The system remains intentionally read-only in the dashboard. It does not dispatch jobs, approve reviews, close Water Emergency records, call vendors, use AI authority, or modify production data from the frontend.

Plain-English Project Status

The project is in a strong foundation position. The main backend structure, database direction, workflow safety rules, read-only dashboard contracts, local testing workflow, and Water Emergency visibility foundation are in place.

The system is not ready yet for production office use because the next major pieces are still future work:

- employee login and permissions
- real Manual Review actions
- real dispatch action screens
- real Water Emergency operating actions
- real vendor connections
- production server deployment
- production backup and security setup

This is the correct order. The system is being built safely so that later operational actions do not bypass human review, dispatch the wrong job, mix Water Emergency work into standard dispatch, or let an external vendor system become the source of truth.

What Each Module Adds

Module 1 - Backend Foundation

Creates the first backend structure for the system. This is the base server that future ACS office tools will connect to. It includes the health check, configuration setup, database migration foundation, and the first production-oriented backend layout.

Module 2 - Core Data Model

Defines the main types of information the system needs to understand, such as customers, properties, jobs, work orders, visits, technicians, Manual Review items, audit logs, and Water Emergency records.

Module 3 - Backend Safety Hardening

Adds safer environment handling so local development, testing, and future production use can be separated. This helps prevent unsafe production settings or accidental secret exposure.

Module 4 - Database Access Boundary

Creates a cleaner separation between database access and business decisions. In plain English, this means the system is organized so that storing information and deciding what should happen are not mixed together.

Module 5 - Intake Checking

Starts the process for receiving job information and checking whether it looks safe, clear, complete, or uncertain. This is the first layer that helps prevent bad information from going straight to dispatch.

Module 6 - Manual Review Queue

Creates the foundation for a Manual Review Queue. This is one of the most important safety systems. If a job is unclear, risky, incomplete, conflicting, or possibly a Water Emergency, it can be held for a human to review.

Module 7 - Dispatch Preparation Logic

Connects the intake checks, confidence checks, and Manual Review preparation into one dispatch-readiness result. It does not dispatch work. It only helps determine if work appears ready, blocked, or review-required.

Module 8 - Intake Record Storage

Stores intake results so the system can keep a traceable history of what came in, what was checked, what was blocked, and why.

Module 9 - Controlled Job Creation

Allows approved standard intake records to become standard jobs in the system. It blocks unsafe, duplicate, review-required, or Water Emergency-separated records from becoming normal jobs.

Module 10 - Work Orders and Visits

Creates the foundation for turning standard jobs into work orders and visits. A work order describes the work to be done, and a visit represents a scheduled field visit.

Module 11 - Assignment and Scheduling Readiness

Prepares information needed for technician assignment and scheduling. This is still preparation only. It does not assign technicians or dispatch work.

Module 12 - Routing and Dispatch Readiness

Checks whether a visit appears ready for routing and dispatch preparation. It protects against missing technicians, missing schedules, blocked visits, or Water Emergency work being pushed into standard dispatch.

Module 13 - Route Assignment and Authorization

Creates the first foundation for route assignment and dispatch authorization. It prepares evidence that a visit is ready to wait for dispatch execution. It does not execute dispatch.

Module 14 - Internal Dispatch Boundary

Adds the internal dispatch execution boundary. This means the system can represent that something has moved into an internal dispatched state, but it still does not call vendors or send anything outside ACS.

Module 15 - External Adapter Preparation

Prepares future payloads for systems such as FastField, Google Sheets, Google Calendar, and technician mobile workflows. It does not call those systems yet.

Module 16 - External Confirmation and Recovery Preparation

Adds the foundation for tracking whether future external work was confirmed, failed, or needs recovery. This prepares for future vendor reliability without creating automatic retries yet.

Module 17 - Operational Event History

Creates an append-only timeline of important operational events. This helps with traceability, debugging, accountability, and future audit review.

Module 18 - Controlled External Execution Boundary

Adds a controlled simulation boundary for external execution. This helps prove the lifecycle can support vendor execution later without actually calling live vendors now.

Module 19 - Dispatch Reconciliation

Adds the foundation for checking whether internal ACS records and external execution evidence match. This prepares the system to detect mismatches later.

Module 20 - Replay and Recovery Preparation

Adds preparation for future recovery workflows when something fails or needs correction. It does not replay work, roll back work, or retry vendor actions yet.

Module 21 - Governance and Approval Evidence

Adds a foundation for recording operator approval and governance evidence before sensitive future actions. This prepares the system for safer management controls.

Module 22 - Accountability and Escalation Preparation

Adds the foundation for escalation and incident-preparation evidence. This helps future managers see when something is serious and needs operational attention.

Module 23 - Dashboard Read Models and API Contracts

Creates read-only dashboard information from the backend. This lets the future admin dashboard show operational summaries without owning the business rules.

Module 24 - Frontend Dashboard Foundation

Creates the first Next.js and Tailwind dashboard shell. This is the first visible admin dashboard foundation. It is read-only and does not perform actions.

Module 25 - Dashboard Visual Quality Pass

Improves the dashboard layout, responsiveness, readability, and status visibility. The focus is making information easier to scan for office users.

Module 26 - Full-Stack Local Dashboard Integration

Connects the frontend dashboard to the backend dashboard API in a local development workflow. This helps verify that the frontend and backend work together safely.

Module 27 - Local PostgreSQL Development Workflow

Documents and prepares the local PostgreSQL database workflow so backend dashboard data can be tested against a real database during development.

Module 28 - Local PostgreSQL Bootstrap and Live Verification

Moves from documentation into actual local database verification where possible. It checks the database, migrations, seed data, backend endpoints, and frontend live dashboard mode.

Module 29 - Synthetic Seed Scenarios

Adds safe fake development data to make the dashboard more useful during testing. These examples are not real customers, not real addresses, and not production data.

Module 30 - Scenario Storyboard

Adds a clearer visual explanation of dashboard scenarios. This helps office users understand whether data represents dispatch-ready work, Manual Review, blockers, recovery, governance, Water Emergency separation, or timeline evidence.

Module 31 - Dedicated Water Emergency Dashboard

Creates a separate read-only Water Emergency dashboard area. This is important because Water Emergency work is not the same as standard dispatch work and should not be hidden inside standard job screens.

Module 32 - Water Emergency Detail and Timeline

Adds read-only detail visibility for an individual Water Emergency record. It can show related job, work order, visit, review, audit, and timeline evidence where available.

Module 33 - Water Emergency Equipment, Visits, and Drying Stage

Adds read-only visibility for Water Emergency equipment context, visit-chain information, and drying-stage context where the existing data supports it. It does not manage inventory or approve field decisions.

Module 34 - Water Emergency Review, Exceptions, and Critical Alerts

Adds read-only visibility for Water Emergency review items, blocker reasons, exception indicators, critical-alert counts, and audit references. It helps operations see risk and review context without adding approve, reject, dispatch, close, or escalation buttons.

How The Pieces Connect

Standard Dispatch Flow

The standard dispatch foundation follows a safe chain:

Intake comes in. The system checks it. Unclear work goes to Manual Review. Approved work can become a job. Jobs can create work orders and visits. Visits can be prepared for assignment, routing, and dispatch. Later modules prepare external vendor evidence, confirmation, recovery, governance, and accountability. The dashboard then displays the current state.

This flow is designed to avoid unsafe automatic dispatch.

Water Emergency Flow

Water Emergency work is intentionally kept separate. Standard dispatch modules repeatedly block Water Emergency records from going through normal job, visit, routing, dispatch, vendor, reconciliation, governance, and accountability paths.

Modules 31 through 34 then add dedicated Water Emergency visibility instead of forcing emergency work into standard dispatch. This protects the special operational nature of Water Emergency jobs, which may involve multiple visits, drying checks, equipment left onsite, pickup, and additional review.

Frontend Dashboard Flow

The dashboard is built on backend-owned read-only information. This means the screen shows what the backend says. The frontend does not invent workflow state, approve actions, dispatch work, or call vendors.

This is important because office dashboards should display reliable operational truth, not create hidden business rules.

What Is Complete Now

The ACS Master System now has:

- a production-oriented FastAPI backend foundation
- a PostgreSQL database and migration direction
- a clean modular project structure
- core operational entities and relationships
- Manual Review Queue foundation
- standard dispatch preparation and evidence flow
- route assignment and dispatch authorization foundation
- external adapter preparation and controlled execution boundaries
- event history and audit-trace foundations
- reconciliation, recovery, governance, and accountability preparation
- read-only dashboard API contracts
- Next.js/Tailwind read-only admin dashboard foundation
- local PostgreSQL development and live-dashboard verification workflow
- synthetic dashboard data for realistic local testing
- Water Emergency summary, detail, timeline, equipment, visit-chain, drying-stage, review, exception, blocker, and critical-alert visibility
- documentation and project memory continuity in the expected project locations

Important Safety Boundaries Still In Place

The dashboard is read-only.

The system does not yet:

- dispatch real jobs from the frontend
- approve or reject Manual Review items
- close or resolve Water Emergency records
- call Google Calendar, Google Sheets, FastField, Verizon Connect, or AI services
- use AI as an authority
- deploy to production
- manage real customer production data

These limitations are intentional. They keep Phase 0 safe while the foundation is being built.

Overall Assessment

The project is moving in the right order. The foundation is strong, safety-focused, and designed for long-term growth.

The most important progress so far is not just that screens and backend pieces exist. The most important progress is that the system now has clear operational boundaries:

- the database is the source of truth
- Manual Review protects uncertain work
- Water Emergency is separated from standard dispatch
- frontend screens do not own business logic
- external systems are future adapters only
- AI is future advisory support only
- operational evidence is preserved for traceability

This gives ACS a safer base for future modules that will eventually add real office actions, authenticated users, production deployment, and live integrations.